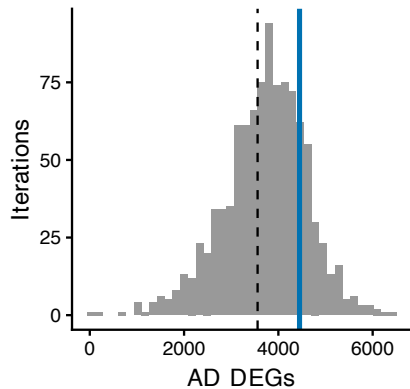
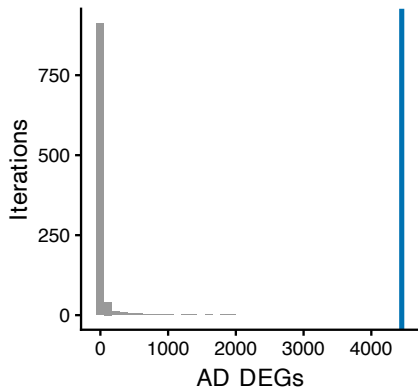


A

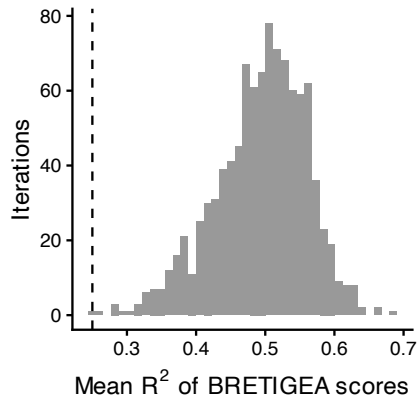
**Arm 1: noise covariates
(retention 0.85)**

**B**

**Arm 2: random-gene PCs
(retention 0.00)**

**C**

**Arm 2: composition captured
(median 0.50)**



Blue line: AD DEGs under Model A (4,449). Dashed lines: pre-specified thresholds.